

Educator creates an account



Create Educator Account

Full Name

Dr. ABC

Institution Email

abc@nus.edu.sg

Institution / Department

NUS School of Computing

Password

.....

Verification

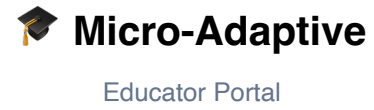
Use institution email to verify educator access before creating courses.

Create Account

[Already have an account? Login](#)

[Next: Create Course →](#)

Educator logs into Dashboard



Email

abc@nus.edu.sg

Password

.....

Login

[Don't have an account? Register here](#)

[Next: Educator Registration →](#)

Course Name

CS101 — Introduction to Python

Course Description

An introductory course covering Python fundamentals including variables, loops, functions and recursion.

Semester

Semester 2, AY2024/2025

Class Size

32

Learning Objectives

- 1. Understand basic Python syntax
- 2. Apply loops and functions to solve problems
- 3. Understand recursion and its applications

Create Course →

Upload Course Materials

Upload your syllabus, lecture slides, and any reading materials. The system will process them into a knowledge base.

Drag and drop PDF / PPT / Excel here
or click to browse files

Uploaded Files

 Week1_Introduction.pdf	 Processed
 Week2_Variables_Slides.pptx	 Processed
 Student_List.xlsx	 Processing...

 No students have joined yet. Generate an invite link and share it with your students.

Generate Student Invite Link

Share this link with your students via email or Canvas. Only enrolled students can join.

`https://t.me/MicroAdaptiveBot?start=CS101-2025-SEC1`

Copy Link

 You can also upload an Excel file with student emails to send invites automatically.

Upload Student List (.xlsx)

Students Joined

0 / 32

Waiting for students to join via the invite link...

Student receives invite and joins Telegram Bot

Student can receive invite via Email / Canvas / Class group chat

NUS Mail

From: abc@nus.edu.sg
To: alice.tan@u.nus.edu
Subject: Join CS101 on MicroAdaptive

Hi Alice,

You have been enrolled in CS101.
Click the link below to join the course Bot
and set up your learning profile:

 **Join CS101 on Telegram**

See you in class!

Clicks link →
opens Telegram

MicroAdaptive Bot

 Welcome, Alice!

I've verified you are enrolled
in CS101 — Introduction to
Python.

No sign-up needed — your
Telegram account is all
you need!

What are your interests?
I'll use these to personalise
your learning 😊

 Gaming

 Finance

 Sports

 Music

 Setup complete! All 32 students have joined. The system is now sending weekly quizzes automatically.

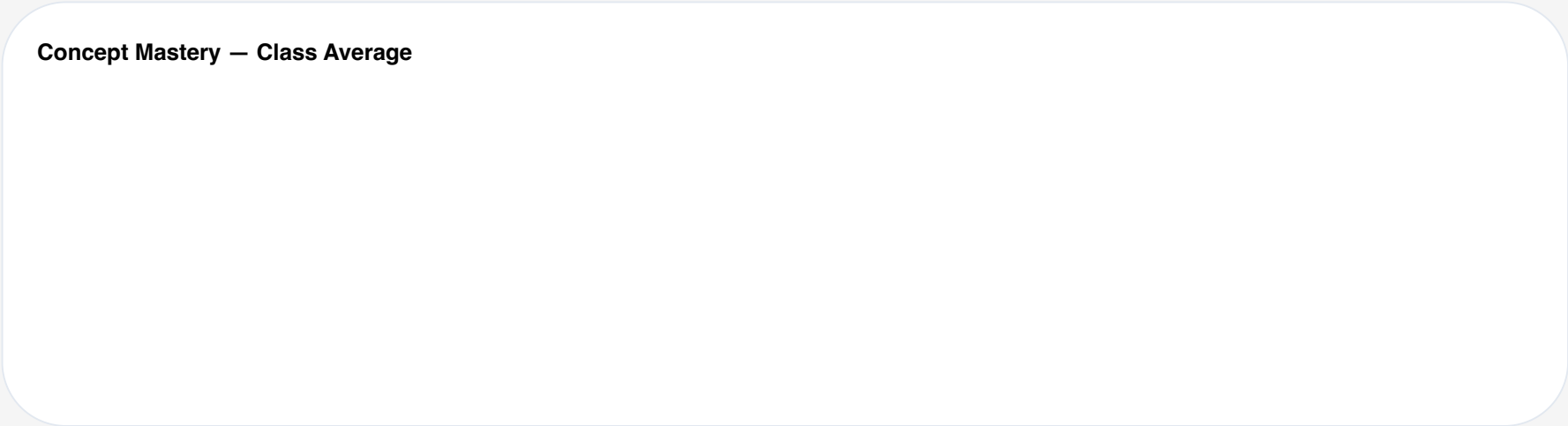
Total Students

32

Class Avg Mastery

Weakest Concept

Active This Week



Setup 1 Login → Setup 2 Create Course → Setup 3 Upload Materials → Setup 4 Generate Invite → Setup 5 Students Join →  System Running

Student receives and answers weekly quiz

① System pushes quiz (Monday morning)

🤖 MicroAdaptive Bot

Good morning Alice! 🙌
Here's this week's question:

What is the output?

```
def foo(n):  
    if n == 0:  
        return 1  
    return n * foo(n-1)  
  
print(foo(3))
```

A) 6

B) 3

C) 0

D) Error

② Wrong answer → Socratic Mode

🤖 MicroAdaptive Bot

B) 3

Not quite! 😊
What do you think happens
when foo calls itself?

it goes back to the start?

Close! Like a game save point.
Each call saves state:

```
foo(3) = 3 × foo(2)  
       = 3 × 2 × foo(1)  
       = 3 × 2 × 1 × 1 = 6
```

③ Mastery updated

🤖 MicroAdaptive Bot

Great work! 🎉
Recursion: 28% → 35% ↑

Variables 75%

Recursion 35% ↑

Keep it up! 💪

Student asks a question (text or voice)

1 Ask via text

 MicroAdaptive Bot

Can you explain recursion?
I don't really get it

Of course! 😊

Since you like gaming, think of recursion like a dungeon in an RPG — each room leads deeper until you hit the final room (base case) and start coming back.

Show code example 🖨

I got it ✅

2 Ask via voice note

 MicroAdaptive Bot

 0:08  

(Whisper API transcribes voice)

I heard your question 🎧

Since you like gaming, think of recursion like an RPG dungeon — each room goes deeper until you reach the final room (base case) and start coming back.

Both text and voice lead to the same personalised response based on student interests

Student views weekly learning progress

① Student checks progress

MicroAdaptive Bot

How am I doing?

Here's your weekly summary 📊

Variables 75%

Loops 66%

Recursion 35% ⚠️

Active streak: 5 days 🔥

Let's keep working on Recursion together 💪

Practice Recursion now

② System pushes weekly summary (auto)

MicroAdaptive Bot

📅 Weekly Check-in!

Here's how you did this week:

✅ 3 quizzes completed

📈 Mastery improved +8%

🔥 5-day streak!

Strongest: Variables (75%)

Needs work: Recursion (35%)

Continue learning 📖

Student updates personal profile (anytime)

1 Request update

MicroAdaptive Bot

update my profile

Sure! What are your interests now? (pick more than one)

Gaming

Finance

Sports

Music

Done ✓

2 Update learning style

MicroAdaptive Bot

Got it! 🎮 Gaming saved.
How do you prefer to learn?

Written explanations

Analogies & examples

Practice quizzes

3 Confirmation

MicroAdaptive Bot

✓ Profile updated!
Your new profile:
🎮 Interest: Gaming
✎ Style: Practice quizzes
I'll use this to personalise your questions and explanations from now on.

Back to learning 🚀

- Overview
- Course Materials
- Teaching Intention
- Students

Total Students

32

Class Avg Mastery

56%

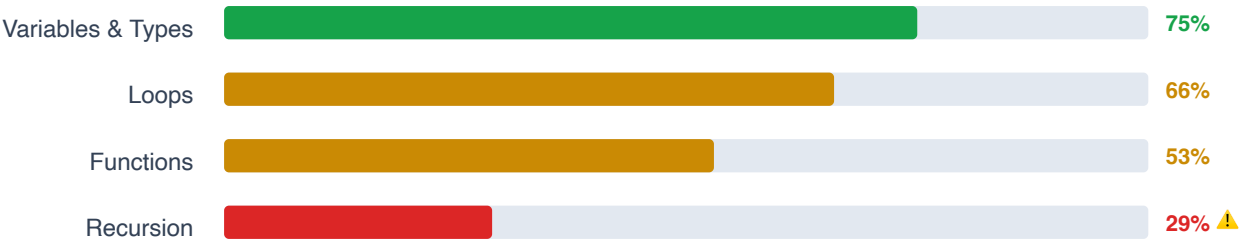
Weakest Concept

Recursion 29%

Active This Week

27 / 32

Concept Mastery — Class Average



💡 Insight: Recursion is the weakest concept — educator should update teaching intention to focus on this.

This Week's Teaching Intention

Focus on Recursion — students are struggling with base cases.
Emphasise the distinction between base case and recursive case.
Use real-world analogies where possible.

Save Intention

Focus Concept

Recursion

Class avg: 29% ⚠ Needs attention

Difficulty Setting

Easy

Medium

Hard

Previous Intentions

Week 3 — Focus on Functions. Practice writing and calling functions with parameters.

Week 2 — Cover Loops. Students should understand for and while loops.

Week 1 — Introduction to Python. Cover variables, data types, and basic I/O.

All Students (32)

Search student...

Alice Tan

61%



Bob Lim ← selected

28%



Charlie Ng

83%



... 29 more students

Bob Lim

Overall Mastery: 28% Needs attention

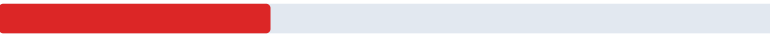
Mastery by Concept

Variables & Types



62%

Loops



35%

Recursion



15%

Recent Activity

27 Jan — Answered Loops quiz. Score: 2/5



26 Jan — Requested explanation for recursion

25 Jan — Answered Variables quiz. Score: 4/5



Learning Profile

Interests: Gaming, Sports

Learning Style: Visual examples

Active Days: 3 / 7 days this week